AEGIS: An Agentic Emergency Guidance and Incident System for Civil Protection

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Abstract. The increasing frequency and severity of climate-driven disasters demands a paradigm shift in how emergency communication systems are designed and deployed. In this paper, we present AEGIS, an agentic platform for civil protection and emergency response on smartphones. The platform integrates a citizen-facing mobile application, a cloud-based backend, an operator dashboard, and an agentic component powered by large language models. The mobile application enables citizens to submit SOS requests, report incidents, receive location-aware alerts, and access evacuation guidance. The agentic component automatically performs risk assessment upon incident detection, combining real-time geospatial, meteorological, and terrain data with information retrieved from the platform's backend to generate and verbalize structured evacuation plans. Recommendations are delivered to operators via the dashboard, where they are reviewed and acted upon under a human-in-the-loop workflow. We evaluate the platform on two dimensions: a usability study of the mobile application and a historical scenario evaluation of the agentic component's risk assessment capabilities. Results demonstrate both the usability of the citizen-facing application and the potential of agentic AI to support time-critical operator decision-making in emergency response.

Keywords: Civil protection · Emergency response · Agentic AI · Large language models · Mobile applications · Human-in-the-loop

1 Introduction

The frequency, severity, and impact of natural disasters have increased sharply in recent years, inflicting loss of life, environmental degradation, infrastructure damage, and major economic costs. Climate change is a central driver, contributing to higher temperatures, prolonged droughts, and more frequent extreme weather events. These conditions intensify wildfires that devastate ecosystems and settlements, while burned landscapes often become more vulnerable to

subsequent floods and landslides. Rapid urbanization further amplifies exposure of large populations to both natural and human-induced hazards. Many disasters also emerge as compound events, where interacting climate drivers converge across spatial and temporal scales, producing cascading and magnified effects. Reducing risk to human life during fast-moving disasters depends on emergency services that can be alerted instantly, provide clear protective guidance, maintain situational awareness, and coordinate rapid assistance to affected areas. To support these functions, civil protection authorities, fire brigades, and police services operate emergency call centers as core components of *Emergency Services Communication Systems* (ESCS). Accessible through standardized numbers, such as 112 in the EU, 911 in the US and Canada, and 999 in the UK, these centers provide universal access to first responders. They allow citizens to report incidents and request aid, while also enabling authorities to disseminate urgent instructions, including evacuation orders, through SMS and other alerting mechanisms.

The ubiquity of smartphones and mobile applications creates new opportunities to strengthen emergency communication and public safety services. Smartphones can use embedded sensors, including GPS, cameras, and microphones, to capture critical contextual information such as precise location and multimedia evidence. This can enhance, validate, and partially automate the information reported by citizens during emergencies. Conversely, authorities can disseminate targeted, timely, and context-aware alerts, including evacuation routes, hazard warnings, safe-area locations, and information on available resources. Such capabilities can improve situational awareness, support safer evacuation from danger zones, and enhance coordinated emergency response. As authorities modernize traditional ESCS infrastructures and transition from circuit-switched systems to Internet Protocol (IP)-based platforms, smartphones and mobile applications are expected to become key interfaces for two-way communication between citizens and emergency services. In parallel, the rapid development of large language models (LLMs) and agentic AI systems creates opportunities to augment human operators with automated reasoning over heterogeneous data sources, including geo-spatial features, meteorological conditions, terrain profiles, and real-time incident reports. These capabilities can support time-critical decision-making during emergencies.

In this paper, we present a platform called **AEGIS** (Agentic Emergency Guidance and Incident System) for providing civil protection and emergency response services on smartphones. The main contributions of this work are: (i) the design and implementation of AEGIS, integrating smartphone-based emergency communication with agentic AI for operator decision support; (ii) a usability evaluation of the citizen-facing application through a scenario-based study with 19 participants, showing high satisfaction and usability across age groups; and (iii) a proof-of-concept of the agentic component on a devastating wildfire in Greece. The remainder of this paper is organized as follows: Section 2 discusses related work; Section 3.1 presents the AEGIS platform architecture and its components; Section 4 presents the evaluation; and we conclude in Section 5 with a summary of findings and directions for future work.

2 Background and Related Work

Since the earliest developments of public warning systems, authorities have sought to design alerts and delivery mechanisms that balance accuracy, clarity, and urgency [24]. As these systems evolve from traditional broadcast methods to mobile communications and social platforms, studies increasingly examine how message features and communication dynamics influence public response during crises [3]. In an early study, the authors of [21] reviewed prior work, identified the stages of a typical warning sequence (hear → confirm → understand → believe → personalize → act) and demonstrated its applicability to both pre- and post-impact contexts [20]. Their Warning Response Model (WRM) stressed that effective warnings must be hazard-specific, provide clear protective guidance, identify who is at risk and where, state when action is required, and cite trusted sources. They also showed that individuals rarely act on the first warning alone, but typically seek confirmation through multiple channels and social contacts, making clarity, consistency, and repetition essential for timely protective action. Public response to post-impact warnings is also shaped by prior experiences of loss, damage, and disruption. Authorities have sought to operationalize these principles in systems like Wireless Emergency Alerts (WEAs) [5]. Early studies showed that adding maps to WEAs only marginally improved message comprehension, with limited effects on compliance to instructions and information s haring [18]. This highlights the need for improved visual aids, such as dynamic and interactive map visualizations, to better support real-time emergency decision-making, particularly because now most users are familiar with smartphones and map-based applications. [2].

Offline social networks, including families, community groups, kinship ties, and voluntary associations, strongly influence how emergency warnings are shared, interpreted, and acted upon [21, 20]. Trust within these networks improves message credibility, encourages protective action, and provides support during crises. As online social networks have become central communication platforms, they create opportunities for disseminating emergency alerts [4, 29, 1] and embedding social communication features into alert systems. However, their effectiveness depends on cultural and community context, while misinformation and disinformation risks make message accuracy and consistency essential.

The application of AI and machine learning to emergency management has attracted growing research attention. Early work focused on mining social media streams during crises to extract situational awareness, detect eyewitness reports, and identify requests for help [13, 22]. More recently, large language models have demonstrated strong capabilities in summarizing incident reports, classifying incoming requests by urgency, and generating natural-language guidance for affected populations [16, 33]. Agentic AI systems extend these capabilities by autonomously selecting and invoking external tools to gather information [23], generate structured outputs, and verbalize recommendations within short time frames. Their use in safety-critical settings, however, raises important design considerations, particularly the need to preserve meaningful human control over automated recommendations. This foregrounds the concept of *human-in-the-loop*

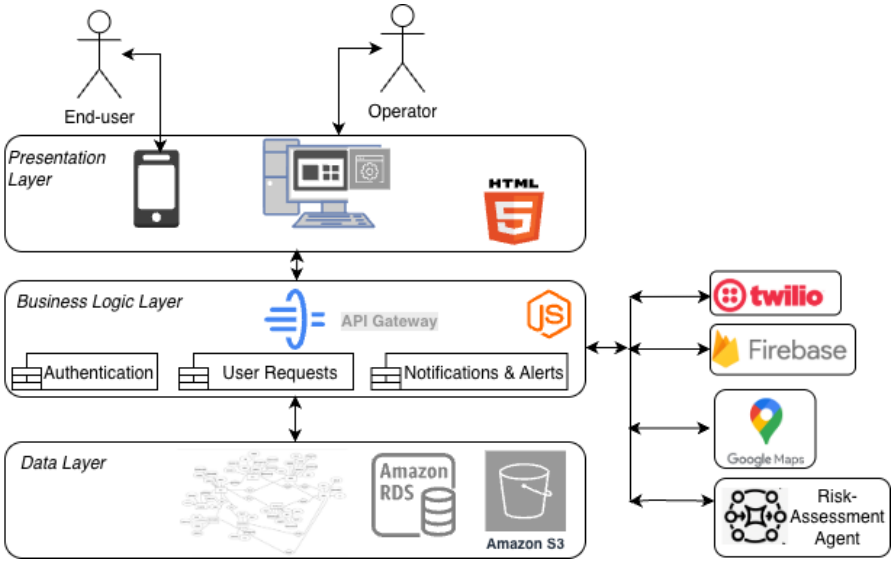


Fig. 1. AEGIS system architecture.

AI, in which automated systems generate candidate decisions that human operators review, modify, and approve before action is taken. Such approaches have been advocated as essential in domains where errors may carry severe consequences [25].

3 Design Concepts and Implementation

3.1 System Architecture and Technology Stack Overview

AEGIS is designed as a framework comprising: i) The **e112 cross-platform smartphone application**, which provides citizens with access to location-aware emergency reporting and disaster-related information. e112 goes beyond the one-way broadcast model of Wireless Emergency Alerts as it supports dynamic interactive map visualizations, location-aware incident reporting, and group communication among citizens and responders. ii) The **AEGIS server**, a back-end system that supports emergency operators through a web-based dashboard, manages context-aware communication with users through the e122 smartphone application and with an agentic layer offering dynamic risk assessment, and interacts with external services for geo-spatial data, synchronization, secure user verification and communication. iii) The **AEGIS Risk Assessment Agent**, which employs a large language model within an agentic tool-use loop to perform multi-source geo-spatial risk assessment and synthesize emergency evacuation plans.

The system architecture of AEGIS follows the principle of separation of concerns, splitting the functionality into independent layers for presentation, business logic, data management, and agentic processing, so that each layer

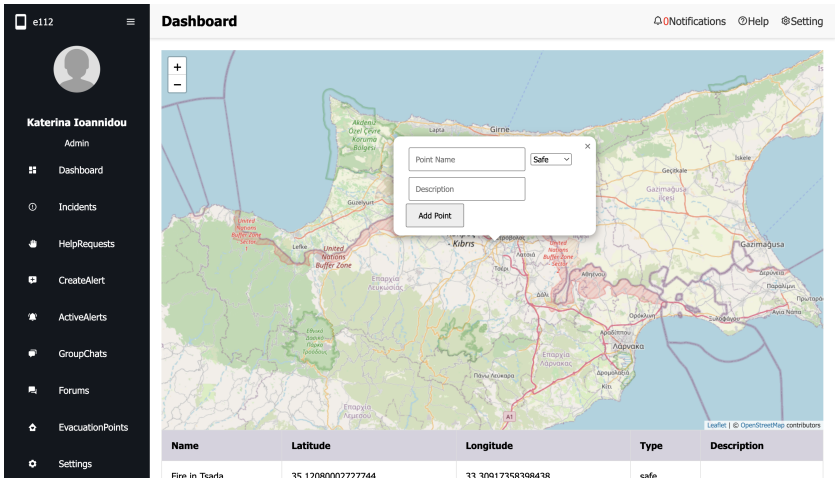


Fig. 2. AEGIS administrator's dashboard: map-oriented interface.

cannot affect another and system modules can be implemented as cloud microservices [27]. This makes the platform easier to scale and troubleshoot.

The **presentation layer** comprises the web dashboard used by AEGIS operators to log in, monitor incoming alerts, manage group communications, and issue official updates (see Fig. 2), and the e112 application, which runs on end-user smartphones and provides reporting and communication functionalities to citizens (see Fig. 3). Both dashboard and app communicate with the back end through AEGIS' API, which is implemented in the business-logic layer. The AEGIS mobile application was implemented with Flutter [10], a software framework for developing native-quality mobile apps across Android and iOS from a single code-base.

The **business-logic layer** comprises the AEGIS server and its API implementation. It is responsible for managing the main operations of the system including authentication, user requests, generation and dissemination of alerts, and communication with the data layer. It handles interactions with the e112 app, the operator's dashboard, and the Risk Assessment Agent. It also integrates with external services: the Google Maps API [11] for location information, the Twilio API [28] for OTP-based SMS verification, and the Firebase API [6] for push notifications.

The **data layer** is responsible for the persistent storage and retrieval of all information required by the system to operate effectively. We designed an entity-relationship model to capture core AEGIS entities and their attributes, including users, alerts, incidents, help requests, chats, and evacuation routes, along with their interrelationships. We implemented this model as a relational PostgreSQL database deployed on Amazon RDS, ensuring reliability and scalability for structured data. Multimedia files such as images and videos are stored externally via Cloudinary, with the resulting media URLs held in the database. The data layer

interacts with the business logic layer through secure, private communication channels, ensuring both efficiency and data integrity.

The **agentic layer** comprises the code of the Risk Assessment agent, along with the local large language models it uses. The agent communicates with the platform exclusively through the business-logic API, with no direct access to the database or the mobile application, and the human operator moderating the submission of information from the agents to the system.

We used the Vercel [30] Platform-as-a-Service (PaaS) cloud to host the presentation, business-logic, and agentic layers. Vercel provides serverless deployment, automatic scaling, and global distribution.

3.2 e112 Smartphone Application Design

We formulated the functionality of *e112* by designing the application's *User Experience* (UX), to ensure that its design takes into account key usability, utility, and emotional impact concerns [14]. To this end, we modeled user needs through *personas* (end users and system operators), *customer journeys* [7], and *empathy maps* [17]. For example, in a wildfire scenario, journey mapping clarified the sequence of actions (receiving an alert, navigating to a shelter), while empathy mapping revealed stress, confusion, and the need for reassurance that could hinder decision-making. Through the UX analysis approach we established and implemented the following key functionalities for the e112 smartphone application: **i) Emergency Alerts' Dissemination**: The system delivers alerts and situational updates to people, in real time, based on their location, thereby minimizing delays and reducing call-center overload during emergencies (Fig. 3a). Upon receipt of an alert, users are able to immediately find simple instructions posted by emergency authorities on what to do (see Fig. 3b,c) or seek help from other people in the area from a dynamically established, localized chat-group (Fig. 3d). **ii) Reporting to and Communication with Authorities**: The system enables users to engage in trusted communication with authorities, requesting assistance under personal duress (Fig. 3e), reporting incidents (Fig. 3f, 3g), getting confirmation receipt (Fig. 3h), monitoring status updates, and obtaining follow-ups (Fig. 3i). The design of the application's graphical user interface and interaction must adapt to the user's anticipated level of stress, with a minimum number of clicks required to ask for help when in personal danger (see Figure 3e,f). Information about the location of the user and/or the reported emergency can be retrieved directly from the smartphone and/or by using its GPS and camera (Fig. 3g). **iii) Information Resources and Presentation**: The application must present information clearly and accurately in visual and textual formats, including annotated maps and landmark images, while minimizing cognitive load under high-stress conditions. The application shall provide intuitive navigation to quickly locate critical resources such as safety guidelines, shelters, protected spaces, hospitals, and police stations (Fig. 3j,k). **iv) Group Communication**: The system allows users affected by the same emergency in the same location to join moderated chat groups that are dynamically established by the platform's operator at the onset of an incident. Group

chat enables individuals to exchange information and advice, help each other even before the arrival of rescue teams, and coordinate activities within a channel monitored by emergency services (Fig. 3d). Moderation and monitoring will ensure that authorities maintain up-to-date situational awareness, are able to deliver targeted guidance and coordinate corrective action remotely, while protecting the integrity of the information exchange from malicious information attacks.**v) Secure Accessibility**: The system facilitates registration through verifiable, non-anonymous credentials (e.g., identifiable mobile numbers or third-party authenticated accounts; see Fig. 3l) to ensure accountability and prevent spam or disinformation attacks. **vi) Interface Responsiveness**: The application adapts seamlessly across devices and screen sizes to ensure a consistent user experience. Additionally, for the graphical user interface (GUI) of the mobile application, we apply a color scheme chosen to balance urgency with usability under stressful conditions, and facilitate navigation decisions under duress.

3.3 AEGIS Server

The AEGIS backend server is a cloud-hosted Node.js/Express application that provides the central communication and data-management layer of the AEGIS platform. It exposes a RESTful API through which the e112 mobile application, the operator dashboard, and the Risk Assessment Agent exchange data and coordinate platform operations.

The backend manages the platform's core operational workflows, including OTP-based user registration and login, user profile management, incident reporting and status tracking, help-request submission and handling, alert creation and dissemination, evacuation-point management, community forum activity, and group-chat communication. The main application state is persisted in a PostgreSQL relational database, including users, incidents, alerts, evacuation points, help requests, group chats, forum content, and agent-generated alert recommendations. Multimedia files submitted through the mobile application are uploaded via the backend to Cloudinary, while the resulting media URLs are stored in the database and reused by the mobile and dashboard interfaces.

The API comprises approximately 90 endpoints organized into nine functional groups. The first group covers *authentication and user management*, including OTP-based registration, login, and profile operations. The second supports *incident management*, including incident submission, status tracking, cluster verification, and operator assignment. The third covers *alert management*, including the creation, editing, deletion, radius-based retrieval, and dissemination of important and non-important alerts, as well as the three-step agent recommendation workflow for submitting, approving, and rejecting AI-generated alert recommendations. The fourth group supports *evacuation and safe-zone management*. This includes the creation, retrieval, and deletion of evacuation points, authoritative safe zones, and operational road closures, together with life-cycle operations for verifying, suspending, and reactivating safe zones. This group also exposes the `official_operational_context` endpoint, which is consumed by

the Risk Assessment Agent to retrieve approved operational data. The remaining groups cover *help requests*; *group communication and messaging*, including dynamic chat groups and operator broadcast messages; *community forum and posts*, including posts, comments, likes, and bookmarks; *safety guidelines and categories*; and *dashboard analytics and media upload*, including incident statistics by region and year and citizen-submitted multimedia evidence.

3.4 Risk Assessment Agent

A central design goal of AEGIS is to reduce the cognitive burden on emergency operators during rapidly evolving, high-impact incidents, when situational awareness is incomplete and decisions must be made under time pressure. To this end, AEGIS includes an agentic component built around a *Risk Assessment Agent*. When a new incident is registered in the AEGIS server, the agent automatically initiates an LLM-driven risk-assessment workflow using the emergency type, location, and contextual notes submitted with the incident.

The agent's data-gathering phase draws on two categories of information. The first uses the `get_official_operational_context` endpoint, which queries the AEGIS backend for platform-held data, including registered evacuation points, authoritative safe zones, and active road closures near the incident. The agent prioritizes this official operational context over external sources to maintain consistency with the platform's current operational picture. The second category consists of *external situational tools*, which enrich the incident with environmental and geo-spatial context, resolving the incident address to geographical coordinates, retrieving nearby roads, vegetation, water bodies, and amenities, current or archived ERA5 meteorological data, and terrain elevation. The LLM decides which external tools to call and in what order, while the agent loop services requests until the model is ready to produce a plan.

The Risk Assessment Agent is implemented in Python as a custom tool-use loop built directly on the Anthropic API, using Claude Sonnet 4.6 as the default model. The implementation uses no agent framework; tool routing, self-correction retries, and structured output capture are handled by approximately 400 lines of application code, with a thin compatibility layer that allows the same loop to run against any OpenAI-compatible backend, including locally-hosted open-weight models via Ollama.

The final output is emitted through a sixth tool, `assess_risk_and_plan`, which functions as a *structured output mechanism* rather than an external service call. Its JSON schema defines the complete *EmergencyPlan*, including risk level, reasoning, affected radius, public alert title and description, evacuation routes, safe zones, affected areas, and immediate actions. The schema constrains the model to populate all required fields and prevents free-form substitutions. Field descriptions also act as grounding instructions: for example, the `road_names` field is annotated as *"road names from OSM data"*, encouraging the model to anchor route recommendations in retrieved geo-spatial evidence. Before accepting a plan, the agent applies three layers of automated validation, checking safe-zone coordinates retrieved from the Open-Meteo elevation API, verifying that no safe

zone lies within the downwind sector during a forest fire, and cross-checking route road names against OpenStreetMap to flag potentially hallucinated names. Any violation is appended to the message history, and the LLM is asked to correct the plan, up to two attempts.

Human-in-the-loop: A core design principle in AEGIS is that the agentic component operates as a *decision-support system*, not as an autonomous actor. When the *Risk Assessment Agent* produces an *EmergencyPlan*, it submits the plan to the backend as a pending alert recommendation. Operators can view pending recommendations through the dashboard and either approve or reject them. Approval converts the recommendation into an active high-priority alert, sends push notifications to mobile application users via Firebase Cloud Messaging, and registers any recommended safe zones and affected areas as official operational entities in the platform database. Rejected recommendations are retained with review metadata. This human-in-the-loop workflow ensures that operators retain full accountability for all directives issued under their authority, while benefiting from the agent's ability to rapidly synthesize heterogeneous data sources that would otherwise require manual lookup.

4 Evaluation

The performance of an emergency application is determined not only by its technical robustness but also by the clarity and efficiency of its user experience, particularly under high-pressure conditions [19]. To examine these dimensions in AEGIS, we evaluated its two main components: (i) the administrator's dashboard, through which authorized personnel receive, manage, and coordinate alerts, and (ii) the e112 application. To this end, we employed a mixed-methods evaluation strategy, combining automated auditing of the administrative dashboard with scenario-based usability testing of the mobile application. This approach was selected to capture both quantitative performance indicators (e.g., page load times, accessibility compliance) and qualitative insights into end-user interaction under simulated emergency conditions.

Administrator's Dashboard: The administrative web interface was assessed using Unlighthouse, an open-source auditing framework that reports on speed, accessibility, code quality, and search engine optimization (SEO) metrics [32]. Using Unlighthouse, we measured the following Core Web Vitals: i) *First Contentful Paint (FCP)*: the time taken to render the first visual element after page load; ii) *Largest Contentful Paint (LCP)*: the time at which the main content becomes visible. iii) *Total Blocking Time (TBT)*: the cumulative duration during which the main thread is blocked after FCP, delaying user interactions. iv) *Cumulative Layout Shift (CLS)*: the extent of unexpected visual shifts during page loading.

Each of these metrics is associated with a recommended threshold for an optimal user experience. Specifically, an FCP $\leq$1.8 s is considered good, while values $>$3 s require improvement. LCP $\leq$2.5 s is desirable, with values $>$4 s negatively affecting usability. TBT should remain below 200 ms to ensure responsive inter-

actions, while a CLS <0.1 is recommended to maintain visual stability during loading. Based on our tests and measurements, the AEGIS dashboard generally performed well. The recorded FCP slightly exceeded the recommended threshold (1.8 s) but remained within the acceptable range (<3 s), possibly reflecting the additional time required for font rendering. LCP was measured at 2.8 s, marginally above the optimal threshold, likely due to the rendering of large text blocks. In contrast, both TBT and CLS achieved ideal values, demonstrating excellent responsiveness and visual stability.

The audit results further indicate a generally high-quality implementation of the administrative dashboard. The system achieved a performance score of 90/100, placing it within the "good" range of the Lighthouse standard [8]. Accessibility achieved 95/100, consistent with WCAG 2.1 guidelines [31]. The Best Practices score reached 96/100, confirming adherence to high standards of security and code quality—both critical for systems that process sensitive emergency data [12]. By comparison, the SEO score was 82/100, an acceptable outcome but one that highlights the need for improvements in metadata and indexing [9].

e112 Mobile Application: For the AEGIS mobile application's usability study, we recruited 19 participants from 16 to 65 years of age to account for known differences in technology use: older adults often adapt more slowly to new interaction patterns whereas younger users typically complete tasks more quickly and efficiently [15, 26]. This approach ensured that the evaluation captured a representative spectrum of usability challenges across diverse user demographics. All participants had prior experience with smartphones and possessed basic digital literacy. For this study, we assumed a simulated flooding disaster scenario to evaluate the application's usability in a high-stress, real-world context. We asked participants to consider they are in the midst of a flood disaster and use the e112 application to submit incident reports and navigate the app's primary features. Following task completion, participants filled a structured questionnaire measuring perceived usability, task difficulty, and overall satisfaction.

We designed the questionnaire to collect meaningful and reliable feedback capturing both quantitative ratings and qualitative perspectives. Our questionnaire used a mix of five-point Likert scales, yes/no questions, and short open-ended items. We asked participants to respond to 16 questions covering a range of aspects of the application: ease of navigation and success in locating desired information; the usefulness of alerts and real-time updates during an emergency; which features they considered most valuable; whether they experienced any difficulties during the reporting process; and how satisfied they were with the response time. We also included items on the ease of posting in the forum, the helpfulness of the evacuation-route tool, the utility of the guidelines feature, and the value of the live chat function. Finally, we asked participants how necessary they believed such an application would be in emergency situations.

After analyzing the responses, we found that participants generally considered the application easy to navigate and reported no major difficulties with the user interface. They also indicated high satisfaction with the response time. An interesting result emerged regarding the live chat feature: participants aged

35–54 viewed it as unnecessary, while those aged 18–34 rated it as highly useful. Overall, we observed high satisfaction with the application, with a mean score of *4.58 out of 5*, and a recommendation score averaging *4.63 out of 5*. Nearly all participants agreed that the existence of such an application is important during emergency situations.

Agent Evaluation We evaluated the risk assessment agent against a major wildfire disasters in SE Europe: the 2018 Mati wildfire (Attica, Greece). This event is documented by post-incident technical investigations and satellite damage assessments, providing ground truth against which agent outputs can be verified. The Mati incident occurred on 23 July 2018,when a wildfire ignited in the Daou Pentelis area of eastern Attica, and spread rapidly eastward toward Neos Voutzas, Mati, and the coastal area of Kokkino Limanaki, leading to more than 100 fatalities. Post-event investigations and case-study accounts attribute the catastrophic loss of life in part to delayed and inadequate evacuation, poor operational coordination, and the entrapment of residents and visitors in coastal and densely built forest-urban interface areas. We configured the agent in historical simulation mode at the point and time of the wildfire's ignition, with meteorological data sourced from the Open-Meteo ERA5 reanalysis archive [34]. The agent retrieved OpenStreetMap (OSM) features within a 3 000 m radius and ERA5 weather data for the archived date and hour. It classified the risk as **critical**. This classification is consistent with the historical record, which describes strong W/WNW winds, very rapid eastward fire spread toward the coast, low humidity near Rafina, and severe evacuation constraints. The agent's risk reasoning explicitly referenced wind direction, fire-spotting potential, and coastal proximity as compounding factors.

The generated evacuation guidance directed evacuees inland and treated Mati beach and the Kokkino Limanaki coastal strip as hazard areas rather than safe zones. Automated geometric validation then checked whether proposed safe zones were outside the downwind hazard sector derived from the incident wind bearing and whether they were sufficiently distant from the coastal entrapment areas identified in the historical record. The validation mechanism triggered once: an initially proposed route toward the coastal strip was flagged and replaced with an inland route to a cleared agricultural area. This self-correction claim is established from the agent execution trace rather than from the historical record.

5 Conclusions and Future Work

In this paper, we presented the design, implementation, and evaluation of AEGIS, a cross-platform mobile application and cloud-based system supporting SOS requests, incident reporting, evacuation guidance, targeted alerts, and moderated community interaction, alongside an operator dashboard for situational awareness and response coordination, and an agentic component for automated risk assessment and evacuation planning under a human-in-the-loop workflow. Our performance assessment and usability studies show that AEGIS is technically

Table 1. Historical fire-scenario evaluation summary.

Scenario	Assessment Summary
Mati, Attica	`-emergency fire -date 2018-07-23 -time 17:00` – CRITICAL risk; correct wind direction: 265° W/WNW; FWI 80/100. – Routes correctly go SW/S, away from the coast and the fire front. – Safe zones are on land: Nea Makri school, fire station, and road junction; they explicitly avoid Mati/Kokkino Limanaki beach. – Affected areas correctly flag the coastline as a death trap, with historical reference. – No sea-coordinate violations or geometry violations. – No tool failures.

robust, highly usable, and well received across age groups. A lightweight, user-centered mobile platform can significantly strengthen preparedness and response, reducing risk to human life during emergencies. Community-driven features, such as contextualized chat forums, further enhance local coordination, while the use of authenticated mobile identities and moderated channels reduces the risk of misinformation attacks. The platform's microservice-based architecture and cloud-native implementation enable scalable deployment, rapid iteration, and seamless integration with third-party services. Our early evaluation of the agentic component on the 2018 Mati wildfire demonstrates that an LLM-based agent, given archived meteorological and geo-spatial data, can reconstruct hazard conditions and generate evacuation guidance that aligns with post-event investigation findings, providing a basis for systematic agent assessment using historical ground truth.

Several interesting open problems remain. A main challenge is the formal evaluation of agentic outputs, requiring not only assessment of final recommendations but also their explainability through trace-level analysis of tool selection, mapping of intermediate reasoning, constraint satisfaction, safety, and correction behavior across realistic emergency scenarios; this aligns with broader concerns that agentic systems require evaluation beyond single input–output correctness. Further work includes automated vulnerability assessment, using the agent to monitor calls for help by citizens and create personalized guidelines, multi-agent coordination across overlapping incidents, integration with legacy emergency services, and AI-assisted moderation to limit misinformation during large-scale crises.

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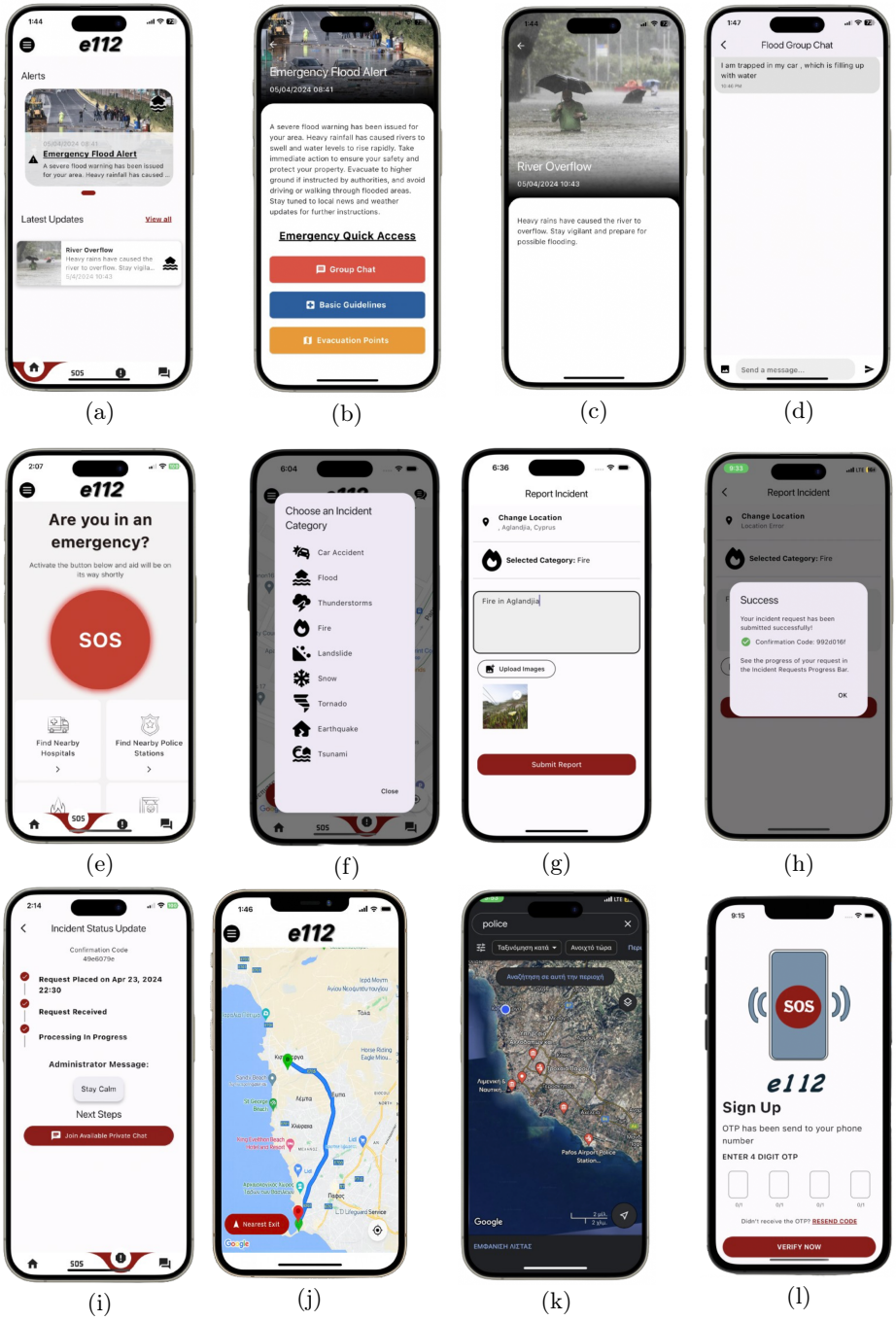


Fig. 3. e112 mobile app functionality and interfaces.